

Parusan Natheeswaran

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SUMMARY OF QUALIFICATIONS

- Computer Engineering student who designs and ships user-centred digital products end to end, from guided multi-step workflows and installable web apps to the APIs and data behind them.
- Designs polished, accessible interfaces, taking ideas from Figma wireframes through prototyping and usability testing with 15+ users to production React front ends built to WCAG standards.
- Works across the full system development life cycle (requirements, design, build, testing, release), with secure JWT authentication, 600+ automated tests, and CI/CD pipelines.
- Clear communicator who works cross-functionally: shipped a feature to millions of players with design, engineering, and QA teams at Riot Games under a 3-month production deadline.

EDUCATION

Toronto Metropolitan University

Sep. 2023 – Present

Bachelor of Engineering, Computer Engineering

Toronto, ON

- **Relevant Coursework:** Algorithms & Data Structures, Operating Systems, Digital Systems, Software Systems, Electronic Circuits I

TECHNICAL SKILLS

Languages: HTML, CSS, JavaScript, TypeScript, Python, SQL, Java, C++, C, Go, Swift, SystemVerilog

Design & Front End: Figma, wireframing, prototyping, usability testing, design systems, WCAG, React, Tailwind CSS

Microsoft 365: Word, Excel, PowerPoint, Outlook, SharePoint, Visio, Power BI, PowerApps, Power Automate

Back End & Systems: Flask, REST APIs, JWT authentication, PostgreSQL, SQLite, Redis, Docker, Windows, Linux, Fly.io

Testing & Delivery: Playwright, Vitest, pytest, test-driven development, GitHub Actions CI/CD

Certifications: Docker Foundations, Atlassian ITSM, GitHub Career Essentials, Generative AI (Microsoft)

WORK EXPERIENCE

Machine Learning Engineer

Oct. 2024 – Present

Outlier AI

Oakland, CA (Remote)

- Enhanced system accuracy and reliability across 20+ AI/ML projects by developing and optimizing speech recognition and conversational AI models using Python, C++, and Java.
- Earned 70+ acknowledgments for improving AI capabilities by leading end-to-end workflows from exploratory data analysis (EDA) to model deployment.

Game Developer

May 2024 – Aug. 2024

Riot Games

Los Angeles, CA (Remote)

- Implemented a new playable character for millions of active users within a 3-month production deadline, collaborating with multiple breakout groups to prototype features and balance mechanics in Python.
- Improved game stability and player experience, cutting crash rates by 15%, by running cross-functional testing and resolving 50+ critical bugs with design, engineering, and QA teams.

PROJECTS

Bingery | React, TypeScript, Python, Flask, Claude API, Fly.io | [Live Demo](#)

Mar. 2026 – Present

- Designed and deployed a full-stack anime discovery platform as an installable PWA, built on a hand-crafted design system (design tokens, glass surfaces, shared motion library) with full reduced-motion support.
- Ran usability testing with 15+ users through surveys and structured feedback, then extensively reworked the mobile interface to fit varied screen aspect ratios (bottom-tab navigation, bottom-sheet modals).
- Built a conversational AI guide with recommend, rate, and onboarding modes using Claude tool-calling, grounded in an 8-signal taste profile and served by a JWT-secured Flask API (72 endpoints).
- Ensured reliability with 619 automated tests written TDD-first (pytest, Vitest, Playwright end-to-end) and daily GitHub Actions jobs keeping a 14,000+ title catalog and sub/dub schedules current.

Xenolinguist | React 19, TypeScript, Electron, Node.js, Ollama, whisper.cpp | [Download](#)

Apr. 2026 – Present

- Designed a guided six-phase decoding workflow with keyboard shortcuts, a Ctrl+K command palette, and confidence scores shown through both colour and type weight; shipped as a v1.0.0 Windows installer.
- Built a fully offline voice pipeline (whisper.cpp, wav2vec2 IPA recognition, espeak-ng) that clearly labels transcriptions versus phonetic guesses, so users never mistake uncertain AI output for fact.
- Added a context-aware AI chat panel and an AI-generated “sandbox” practice language with Easy/Medium/Hard levels, running on streaming local Ollama models with nothing leaving the user’s device.

Pulsar-V | SystemVerilog, RISC-V, FPGA, cocotb, SymbiYosys, Yosys/nextpnr

Jul. 2026 – Present

- Designed two open-source RV32I CPU cores in SystemVerilog and validated them against industry standards (43/43 riscv-formal checks, 38/38 riscv-arch-test, 41/41 riscv-tests), with fully documented results.